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14	C	From graph, $x_0 = 0.30 \text{ m}$, $T = 5.0 \text{ s}$, $\omega = \frac{2\pi}{T} = \frac{2\pi}{5.0} = 1.26 \approx 1.3 \text{ rad s}^{-1}$ Displacement of pendulum, $x = x_0 \sin(\omega t)$ Velocity of pendulum, $v = \frac{dx}{dt} = \omega x_0 \cos(\omega t)$ $= (1.26)(0.30) \cos(1.3t)$ $= 0.38 \cos(1.3t)$
15	A	When the volume control of the radio is raised, there is no change to the